Note:
Synchronous simply means that there is a precise time relationship between the two frequencies and input binary bit rate.
The mark and space frequencies are selected such that they are separated from the center frequency by an exact odd multiple of one half of the bit rate.

Mark or Space Frequencies

$$f_m \text{ and } f_s = n\left(\frac{f_b}{2}\right)$$

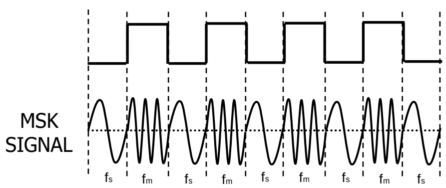
where:

f_m and f_s = mark and space frequencies

n = any odd integer

This ensures that there is a *smooth phase transition* in the analog output signal when it changes from a mark to a space frequency or vice versa.

MSK Waveform



Advantage:
Has a better BIT-ERROR Performance than the conventional binary FSK for a given signal-to-noise ratio.

Disadvantage:
It requires synchronizing circuit and is therefore expensive.

Gaussian Minimum Shift Keying (GMSK)

Used in the GSM cellular radio and PCS systems. The word Gaussian refers to the shape of the filter that is used before the modulator to reduce the transmitted bandwidth. GMSK uses less bandwidth than conventional FSK.

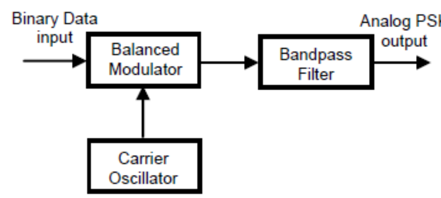
Phase Shift Keying (PSK)

The phase of the analog carrier is varied by the digital modulating pulse. Another form of angle modulated, constant envelope digital modulation. Similar to Phase Modulation except that the input is a binary digital signal. When higher data rates are required in band-limited channels that are capable with FSK, phase-shift keying is often used.

Binary PSK (BPSK)

- Two output phases are possible for a single carrier frequency.
- Another term is **Phase Reversal Keying** (PRK)
- One output phase represent a logic 1 & the other a logic 0.
- A form of suppressed carrier, square-wave modulation of a continuous wave (CW) signal.

BPSK Transmitter



BALANCED MODULATOR - acts like a phase reversing switch. Depending on the logic condition of the digital input, the carrier is transferred to the output either in-phase or 180º out-of-phase with reference to carrier oscillator.

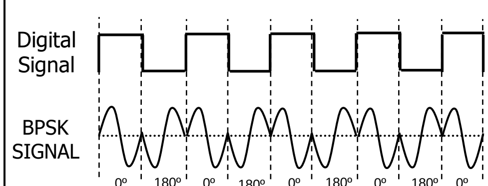
Bandwidth of BPSK Signal

$$f_N = 2\left(\frac{f_b}{2}\right) = f_b$$

Output Baud Rate

$$\text{Baud rate} = f_N = f_b = \text{input bit rate}$$

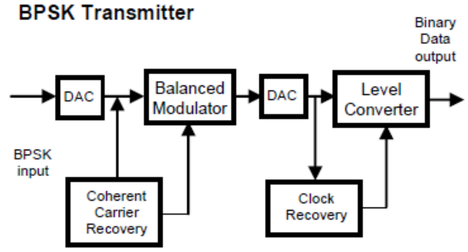
BPSK Waveform



Note that the output signal is either in phase (0º) with the reference oscillator or out-of-phase by 180º.

Binary Input	Output phase
0	180º
1	0º

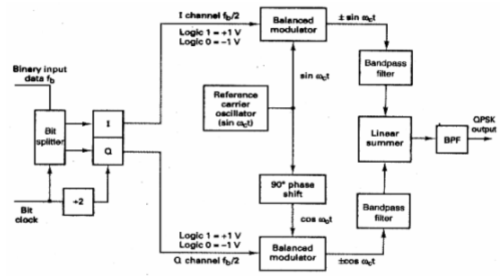
BPSK Transmitter



Quaternary PSK (QPSK)

In **QPSK**, there are four possible output phases for a single carrier frequency. The binary input data for QPSK are combined into groups of 2 bits called *dibits* before being acted by the modulator. For every 2 bits at the input, a single output change is phase angle is occurring thus the output rate of change (baud rate) is one-half of the input bit rate (bit rate).

QPSK Modulator



Minimum Bandwidth Required (f_N)

$$f_N = 2\left(\frac{f_b}{4}\right) = \frac{f_b}{2}$$

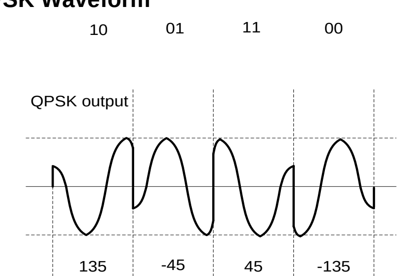
Output Baud Rate

$$\text{Output baud rate} = f_N = \frac{f_b}{2}$$

M-ary Encoding (For QPSK, M = 4)

$$N = \log_2 M = \log_2 4 = 2 \text{ bits}$$

QPSK Waveform



Binary Input		Output Phase
Q	I	
0	0	-135º
0	1	-45º
1	0	135º
1	1	45º

Offset QPSK (OQPSK) or Offset-Keyed QPSK (OKPSK)

This is a modified QPSK where the bit waveforms on the I and Q channels are offset or shifted in phase from each other by ½ of a bit time.